

Validity:

1) **Validity** is a property of arguments and is defined as follows:

An argument is valid **if and only if** it is logically impossible for the premises to be true and the conclusion to be false.

To determine validity, it is absolutely irrelevant whether the premises are **actually** true or false. You classify an argument as valid in the case where, assuming the premises to be true, the conclusion **necessarily follows**.

Examples of valid arguments:

- If it is a dog, then it is a mammal.
 - It is a dog.
 - **Conclusion:** It is a mammal.
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- Either the cake is chocolate or vanilla.
 - The cake is not chocolate.
 - **Conclusion:** The cake is vanilla.
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- All birds have feathers.
 - Tweety is a bird.
 - **Conclusion:** Tweety has feathers.

- All mammals are cool
- All cool things are for young people
- **Conclusion:** All mammals are for young people

- If it is sunny, then I will go to the beach
- I am not going to the beach
- **Conclusion:** It is not sunny

- If the match is lit, then there is oxygen in the room
- There is no oxygen in the room
- **Conclusion:** the match is not lit.

Fill in the conclusion:

- 1) If I am a Greek citizen, I hold a Greek passport

I am a Greek citizen

Conclusion:

- 2) All htg's are fd's

This is a htg

Conclusion:

- 3) If I am a Greek citizen, I am an American citizen

I am a Greek citizen

Conclusion:

4) If the moon is made of cheese, then astronauts can eat it.

The moon is made of cheese.

Conclusion:

5) Either Napoleon was a philosopher or Napoleon was a robot.

Napoleon was not a philosopher.

Conclusion:

6) All fish can fly.

Salmon is a fish.

Conclusion:

7) If squares were triangles, then they would have three angles

Squares don't have three angles

Conclusion:

Logical Symbols:

If the actual meaning of the declarative sentence is irrelevant for determining validity, then it is easier to manipulate symbols instead of sentences. The symbols would represent the fact that there is a sentence/proposition.

Examples:

(P) $\rightarrow$ (Q)

- If it is a dog, then it is a mammal.

(P)

- It is a dog.

(Q)

- It is a mammal.

This tells you that if you have a conditional $P \rightarrow Q$ **and** P , you are allowed to infer Q because it follows necessarily. From the perspective of formal validity, it cannot be the case that both $P \rightarrow Q$ **and** P are true and Q false.

(P) $\rightarrow$ (Q)

- If the match is lit, then there is oxygen in the room

Not-Q ($\sim Q$)

- There is no oxygen in the room

Not-P ($\sim P$)

- The match is not lit.

(P) OR (Q)

- Either the cake is chocolate or vanilla.

(Not-P) ($\sim P$)

- The cake is not chocolate.

(Q)

- **Conclusion:** The cake is vanilla.

This tells you that if P or Q is true and Not-P is the case, then you are allowed to infer that Q is the case. Any other solution is impossible.

Classical Logic Truth-Tables:

Truth tables let you decide the validity of an argument in a more efficient manner.

1) Conjunction: **P and Q, symbolically written $P \wedge Q$.**

P	Q	P AND Q ($P \wedge Q$)
T	T	T
T	F	F
F	T	F
F	F	F

F	F	F
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2) Disjunction: **P or Q**, symbolically written **$P \vee Q$** .

P	Q	P OR Q ($P \vee Q$)
T	T	T
T	F	T
F	T	T
F	F	F

3) Implication: **P implies Q**, symbolically written **$P \rightarrow Q$** .

P	Q	If P then Q ($P \rightarrow Q$)
T	T	T
T	F	F
F	T	T
F	F	T

F	F	T
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Table (3) explains two main inference rules, modus ponens and modus tollens.

Modus Ponens = $P \rightarrow Q$

$$\frac{P}{Q}$$

Modus Tollens = $P \rightarrow Q$

$$\frac{\sim Q}{\sim P}$$

Modus Ponens: The truth table tells you in line that if **P** and **$P \rightarrow Q$** are true, then **Q** is also true. There is no other line where this is the case.

Modus Tollens: The truth table in the last line tells you that if **Not-Q ($\sim Q$)** and **$P \rightarrow Q$** are true, then **$\sim P$** is also true.